Interview Transcript – Macarascas Clinic

Diagnostic Needs and Limitations

(00:17)

Interviewer: Ano po yung pinaka urgent or pressing needs po dito na na-encounter nyo sa pagda-diagnose ng respiratory diseases?

K1_03: Kami po, hindi nagda-diagnose—doktor po ang gumagawa nun.

KI_03: For check-ups, meron kaming stethoscope, pero wala kaming x-ray.

KI_03: Basic lang po talaga dito sa clinic. Kung hindi kaya, referral agad sa hospital.

Teamwork and Workflow

(01:55)

KI_03: Teamwork po kami. Usually 2–3 ang naka-duty. Hati-hati sa tasks.

KI_03: May triage kami—pagdating ng pasyente, kinukuha agad ang vitals, tapos forward sa amin.

KI_03: Kami ang nagsusulat ng complaints, tapos diretso na sa doktor.

KI_03: Kailangan mabilis, kasi kung mabagal, aabutin ka ng gabi.

KI_03: May specific task assignments para efficient ang flow.

Referral Challenges

(03:20)

Interviewer: Ano po yung mga challenges nyo pag nagpaparefer sa hospital?

KI_03: Una, capacity. Apat lang ang hospital—tatlo private, isa public.

KI_03: Sa private, problema agad ang financial aspect.

KI_03: Karamihan dito sa poor sector, kaya sa public hospital kami umaasa.

KI_03: Pero napupuno rin ang provincial hospital, lalo na sa peak season.

KI_03: Kapag puno ang private, tinatapon nila sa ONP.

KI_03: ONP, being government, tatanggapin kahit puno—pero hindi agad maaasikaso.

(04:37)

KI_03: Hangga't kaya sa satellite level, minamanage namin

. **KI_03:** Pero kung critical, wala nang choice—takbo na sa ONP.

KI_03: ONP can't refuse, pero hindi rin agad maasikaso dahil sa dami ng referrals from other municipalities.

Community Education and Preventive Efforts

(05:49)

Interviewer: Paano nyo po na-address ang community misconceptions?

KI_03: Sa barangay health centers, sila ang preventive side—kasama ang IEC.

KI_03: Dito sa amin, kapag admitted na, part na ng management ang education.

KI_03: Pagdating nila dito, may sakit na sila—so hindi na preventive, kundi curative na.

Device Design Feedback

(06:49)

Interviewer: Nagde-develop po kami ng screening or pre-diagnostic device.

Interviewer: Paano po dapat i-design ang device para madaling linisin at i-maintain sa rural clinics?

KI_03: Dapat waterproofed.

Device Hygiene and Infection Control

(07:11)

KI_03: Hindi mo alam kung anong klaseng pasyente ang darating—lalo na sa respiratory cases. May TB patients na sumusuka ng dugo.

KI_03: Dapat waterproof ang material. May plema, dugo—hindi mo alam.

KI_03: Mahirap kung absorbent ang material. Hindi puwedeng i-autoclave ang microphone. Hindi rin puwedeng hugasan.

KI_03: Dapat rust-resistant. Basic alcohol lang ang disinfectant.

KI_03: Mas okay kung may plastic cover—parang sa ultrasound probe. Pero dapat walang effect sa sound quality.

KI_03: Infection control ang priority sa medical devices.

Interface and Training Needs

(09:19)

Interviewer: How important is the interface for nurses with limited digital training?

KI_03: Pwede ba siya sa cellphone?

Interviewer: May microcontroller po sa loob, tapos wirelessly connected sa website.

KI_03: Maganda kung app-based. Mas accessible.

KI_03: Pero dapat may video guide—lalo na sa staff na hindi tech-savvy.

KI_03: Hindi lahat makaka-attend ng training. Sa amin, walo kami pero hindi sabay-sabay.

KI_03: Mas okay kung may user guide sa app.

KI_03: Younger staff prefer reading, pero older staff hindi na nagbabasa.

Integration into Patient Care

(11:53)

Interviewer: How can the device be integrated into current patient care routines?

KI_03: Hindi siya burden. Part ng basic assessment ang auscultation—lalo na kung may ubo o sipon.

KI_03: Mas maganda kung may mic na nagre-record ng sound.

KI_03: Helpful kung puwedeng i-download at i-send sa doktor via Messenger—lalo na kung walang doktor on-site.

KI_03: Pwede sa telemedicine. May mga digital stethoscope na ganito, pero mahal.

KI_03: Sana may option to download or share via link—parang iCloud.

KI_03: Kahit link lang, para ma-share ang sound file.

Sound Interpretation and AI Limitations

(14:02)

KI_03: Sound ang importante—crackles, wheeze.

KI_03: Doktor ang mas particular sa tunog.

KI_03: Machine may error risk.

KI_03: Kahit walang interpretation, basta ma-record ang tunog, kaya na naming i-interpret.

Model Testing and Diagnostic Boundaries

(15:36)

Interviewer: Ito po ang main design at testing ng model.

KI_03: May AI na nag-analyze gamit ang dataset from credible sources.

KI_03: Pero mabigat ang salitang “diagnosis.”

KI_03: Dapat “early warning” lang—para ma-refer agad sa doktor.

KI_03: Halos magkakapareho ang lung sounds ng pneumonia, asthma, etc.—kailangan pa rin ng x-ray.

(16:34)

Interviewer: Model shows confidence level based on 20,000 audio recordings (5 seconds each).

KI_03: Machine compares new audio to dataset and outputs most similar result. **(17:12)**

Interviewer: Output is automatic—e.g., asthma.

KI_03: AI analyzes and suggests possible diseases related to wheezing.

KI_03: Pero hindi dapat umasa agad sa diagnosis.

KI_03: Minsan may wheeze and crackles combined—iba’t ibang interpretation.

Sound-Based Diagnosis and Suggestive Output

(17:21)

KI_03: AI ang nag-a-analyze. Pero dapat naka-encapsulate lang yung output—suggestive of possible diseases related to wheezing.

KI_03: Hindi dapat agad i-diagnose as asthma. Minsan may wheeze at crackles combination.

KI_03: May upper and lower lung sounds—iba-iba. Minsan may “no lung sound” kung puno ng tubig or hangin.

(18:53)

KI_03: Tama rin na “respiratory audio classifier” ang tawag—hindi “disease classifier.”

KI_03: Pwede lumabas na “wheeze sound detected, may be suggestive of asthma.”

KI_03: COPD, asthma, etc.—magkakapareho ang tunog. Kaya dapat suggestive lang.

(19:35)

KI_03: Maganda ang idea, pero mabigat kung “diagnosis” ang label. Parang ECG—hindi nagsasabi ng sakit, kundi rhythm lang.

KI_03: Kapag abnormal ang rhythm, saka ka pa-evaluate.

Interface Suggestions and GUI Accessibility

(20:25)

KI_03: Suggestion: panel outputs like “wheeze detected,” “no sound,” etc.

KI_03: Kahit isa lang ang ma-detect, okay na.

KI_03: Output like “wheeze sound detected, may suggest asthma, please consult…” is helpful.

(21:16)

KI_03: Sound detection is key—whizz, crackles, etc.

KI_03: Whizz = asthma. Crackles = parang tunog ng hair scratch.

KI_03: GUI is helpful—buttons should be clear.

KI_03: It’s another way to confirm findings.

Workflow Integration and Contamination Control

(22:49)

KI_03: Pwede bang iwan lang ang stethoscope sa pasyente?

KI_03: Traditional method requires presence.

KI_03: Pwede ipatong lang, pero kailangan pa rin i-guide ang pasyente.

KI_03: Machine can compile 5-second audio chunks and compute results. **(23:49)**

KI_03: Mas okay kung less contact—lalo na sa infectious cases.

KI_03: Parang sa COVID, may patch-style stethoscope.

KI_03: Pwede rin removable or detachable parts—para lusot sa contamination risk.

(25:20)

KI_03: Material ng vest dapat waterproof—like nylon.

KI_03: Pwede i-connect sa cords, detachable setup.